Padraig Lavery

Backend Software Engineer – Node.js, AWS, PostgreSQL

Belfast, Northern Ireland | LinkedIn | GitHub

PROFILE

Backend software engineer with 4+ years building and running high-volume, security-sensitive services for UK and Middle East banking clients, using Node.js, .NET/C# and PostgreSQL on AWS and Kubernetes. Known for production reliability: fixes root causes, makes batch jobs fail safely, and leaves runbooks so problems don't come back. Leads cross-team engineering work and mentors new starters.

EXPERIENCE

Developer | Reward, Belfast | June 2022 – Present

Backend and production reliability

- Eliminated recurring daily out-of-memory crashes in a high-volume transaction ingestion service by re-architecting cache loading to run once per batch instead of per file; mitigated further memory-exhaustion failures in ETL jobs.
- Redesigned a month-end transactional invoicing batch with per-partner error isolation, explicit timeouts and safe retry/reconciliation, so failures on the highest-volume partners no longer abort the whole run.
- Resolved a live customer-facing outage in a Kubernetes/Istio ingress setup and permanently corrected the Helm deployment templates.
- Authored runbooks and SOPs so the production services team resolves recurring issues without engineering escalation.

APIs, security and platform

- Hardened authentication and account services: removed stack traces and SQL details from error responses and standardised input validation across partner microservices.
- Designed and implemented audit logging of staff access to customer accounts across UK and Middle East bank portals.
- Authored the AWS SDK for JavaScript v2-to-v3 migration guide across Lambdas, Fargate services and microservices; planned testing for Node 22 runtime upgrades and remediated critical Snyk-flagged vulnerabilities for PCI DSS.
- Automated scheduled file processing with EventBridge, Lambda and Fargate; audited 100+ S3 buckets and applied lifecycle policies.

Delivery and leadership

- Led a cross-team initiative for automated API testing: request validation, shared error handling, test-data cleanup endpoints, OpenAPI/Swagger documentation, unit tests and stabilised GitHub Actions CI.
- Built full stack features including a secure e-code resend flow with audit-logged resends and multi-bank email routing.
- Mentored and onboarded four new starters; run the team's retrospectives and author technical guides.

Intern Software Engineer | Flexera, Belfast | May 2020 – April 2021

- Developed a web API for calculating cloud costs using Go, Docker, Kubernetes and Git.

PROJECTS

Incident Copilot (in progress) | Python, FastAPI, PostgreSQL + pgvector, Ollama, Docker, GitHub Actions

- Incident-triage service that retrieves runbook content and suggests likely root causes and next steps, with agentic tool-calling for live service status.

SKILLS

- **Languages:** JavaScript (Node.js), C#, Python, Go, SQL
- **Back end:** Node.js, .NET Core, FastAPI, Flask, microservices, REST and GraphQL APIs, OpenAPI/Swagger
- **Data:** PostgreSQL (incl. pgvector), MongoDB
- **Cloud and infrastructure:** AWS (Lambda, Fargate, S3, EventBridge), Kubernetes, Helm, Istio, Docker, Terraform, GitHub Actions
- **Practices:** production incident response, runbooks/SOPs, automated testing, security remediation (Snyk, OWASP), Agile/Scrum

EDUCATION AND CERTIFICATIONS

- BSc (Hons) Computer Science, 2:1 – Ulster University, Belfast, June 2022
- AWS Certified Developer – Associate; AWS Certified Cloud Practitioner
- AWS Certified AI Practitioner – preparing, exam November 2026